



CHANGE PLAN

VIRTUAL CONTROL

VMWARE CLOUD FOUNDATION SOLUTIONS

Nested ESXi Host CPU/RAM Rebalance

Right-sizing the four nested hosts after the Automation over-provisioning assumption — measured targets, hard constraints, and a safe maintenance-window procedure.

RIGHT-SIZING

CAPACITY

VSAN MAINTENANCE MODE

HA / DRS

CHANGE PLAN

v1.1

VMware Cloud Foundation 9

Nested ESXi Host CPU/RAM Rebalance Plan

Prepared by: Virtual Control LLC **Date:** June 8, 2026 **Document Version:** 1.1 **Classification:** Internal

Scope. The four nested ESXi hosts were sized large on the assumption that the Automation stack would need most of the box's CPU and memory. With Automation now deployed and its real footprint measured, this plan right-sizes the hosts: it relieves the one hot host (esxi03 at ~93% RAM), evens the load, separates the heavy management VMs across hosts, and shrinks the HA failover reserve — all within the physical ceiling and without depending on vMotion (which is currently unreliable on the degraded nested storage).

Why now. On 2026-06-05, DRS (then Fully Automated) tried to live-migrate the vCenter VM off the overloaded esxi03 and the migration crawled/failed on the slow storage, cascading into an HA-agent outage. Right-sizing removes the pressure that caused that.

Status. Proposal for review — **no changes made.** Targets in §4 are a starting point to validate in §5 before execution.

Update — 2026-06-08 (post-execution). This plan was executed via Approach B (full maintenance window). The host resize applied as designed (esxi01 96 / esxi02 72 / esxi03 88 / esxi04 96 GB). During the bring-up, **vMotion was verified working again** — `idm` and `vrs1cm` live-migrated cleanly past the old ~56% stall point with zero downtime, and vSAN was healthy (resync 0). So the **"vMotion unreliable" basis for Constraint #4 and Approach A has cleared**; live vMotion is now usable for placement moves (the cold-only guidance below is kept as history). DRS remains **Partially Automated** pending a longer reliability window before considering Fully Automated. *Note: on power-on the light VMs scattered onto esxi04 (DRS auto-placement, no rules yet) — corrected with live vMotion + soft VM-Host should-run rules; see the Execution Worksheet.*

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1. Goal

- **Relieve esxi03** (currently ~93% RAM — vCenter + NSX Manager both landed on it).
- **Even RAM** across the four hosts so no host runs hot at rest.
- **Separate the three heavy VMs** — `aria` (~54 GB), `nsx-manager` (~42 GB), `vcenter` (~24 GB) — onto **three different hosts**.
- **Shrink the HA failover reserve** (currently 34% CPU / 25% RAM — auto-sized to the largest host; more even hosts = smaller reserve).
- **Stay within the 384 GB physical ceiling** and **do not depend on vMotion** (it can't reliably converge on this storage today).

2. Current Measured State

Physical host: Dell Precision 7920 · 2× Xeon Gold 6140 (36 cores / 72 logical) · **384 GB RAM** · VMware Workstation 17.

Nested hosts (configured vs consumed, measured 2026-06-08):

Host	vCPU	CPU use	RAM cfg	RAM used	RAM %
esxi01	16	10%	88 GB	50 GB	57%
esxi02	12	15%	64 GB	38 GB	59%
esxi03	12	58%	88 GB	82 GB	93% ← hot
esxi04	32	21%	112 GB	83 GB	74%
Total	72	—	352 GB	253 GB	—

Management VMs (configured / host-RAM used / host):

VM	vCPU	RAM cfg	RAM used	Host
aria	24	54	54	esxi04
nsx-manager	6	48	42	esxi03
vcenter	6	24	24	esxi03
vcf-ops	4	20	18	esxi02
idm	8	16	13	esxi01
sddc-manager	4	16	16	esxi01
fleet	6	12	5	esxi02
vrslcm	2	6	6	esxi01
vCLS ×2	1 ea	~0	~0	esxi02/04

The root of the imbalance: `vcenter` (24) + `nsx-manager` (42) both sit on esxi03's 88 GB → 93%. Spreading those two apart is the real fix; host resizing supports it.

3. Hard Constraints

1. **Physical ceiling = 384 GB.** Nested hosts total **352 GB** configured today; the Windows host + Workstation overhead needs the remaining ~32 GB. **Keep total nested RAM ≤ ~352 GB** — rebalance *within* the budget; do not grow the total.
2. **vSAN overhead per host.** Each nested ESXi needs RAM for ESXi + vSAN (ESA/OSA metadata/cache). Don't size any host below **(its VMs + ~20% headroom + vSAN overhead)**. Practical floor ≈ 64 GB for a host carrying a heavy VM.
3. **Resizing a nested host requires powering it OFF** — vCPU/RAM reduction cannot be done live on the nested-ESXi VM in Workstation.
4. **vMotion is currently unreliable** on the degraded nested storage (the 2026-06-05 vCenter vMotion failed to converge). **Do not assume live evacuation works** — plan for cold (powered-off) VM handling. *(Superseded 2026-06-08 — see the post-execution update note near the top: vMotion has since been verified working. Retained as the constraint that was true at planning time.)*
5. **vSAN data availability.** Any host taken down must leave vSAN able to serve data (FTT). Use the correct **maintenance-mode data-migration option** (see §7) — a wrong choice previously caused a vCenter journal abort.
6. **Fragile, interlocked single-host lab** — one change at a time, verify vSAN health before/after each, keep the management appliances powered off (not just MM) during the resize of the host they run on.

4. Target Sizing & Placement (Proposal)

Within the 352 GB budget, even the hosts and split the heavy VMs three ways:

Host	RAM now → target	vCPU now → target	Intended heavy/anchor VMs	Projected use
esxi01	88 → 96	16 → 16	nsx-manager + sddc-manager	~58 GB (60%)
esxi02	64 → 72	12 → 12	vcf-ops + fleet	~24 GB (33%)
esxi03	88 → 88 (keep)	12 → 12	vcenter + idm + vrs lcm	~43 GB (49%)
esxi04	112 → 96	32 → 24	aria	~56 GB (58%)
Total	352	72 → 64	—	even, no host >60%

Rationale: - **aria** stays on esxi04 (its pinned host) but esxi04 trims 112→96 GB and 32→24 vCPU (aria itself runs fine at 24 vCPU; 32 was the over-provision). Frees 16 GB. - **nsx-manager** moves off esxi03 onto esxi01 (grown 88→96 to fit it). This is the key move that relieves esxi03. - **vcenter** stays on esxi03, now nearly alone → drops from 93% to ~49%. - esxi02 grows slightly (64→72) for comfort; small VMs spread across it and esxi03.

Targets are a **validated starting point**, not gospel — confirm each host's projected use (VM + ~20% + vSAN overhead) in §5 before committing. If real vSAN overhead is higher than assumed, bump the affected host and trim esxi02 to stay ≤352.

5. Pre-Flight Checks (Read-Only)

Before any change:

1. **vSAN object health = healthy, 0 inaccessible:** `esxcli vsan debug object health summary get` (per host) — must be clean **before** starting.
2. **No active resync:** vSAN → Monitor → Resyncing Objects = 0.
3. **Confirm overhead assumptions:** for each host, current `RAM used - Σ(VM used on it)` ≈ ESXi+vSAN overhead; use it to validate the §4 targets fit.
4. **Confirm DRS = Partially Automated** (so nothing auto-moves mid-change) and note current VM→host placement.
5. **Back up / export** the management VM inventory list and the current host sizes (so rollback is known).
6. **Pick the window:** all management VMs will be **down**, so vCenter UI is unavailable during the work — drive from the **ESXi Host Client** + Workstation.

6. Execution Approach

Two ways to do this; the second is recommended here.

Approach	How	Verdict
A. Rolling (one host at a time, cluster stays up)	Per host: vSAN MM (data migration) → evacuate VMs by vMotion → resize → return	 Not viable now — depends on vMotion, which is unreliable on this storage; esxi03 holds vCenter which won't migrate
B. Full maintenance window (whole stack down)	Graceful shutdown of all mgmt VMs + vSAN cluster, resize all four hosts cold, then start up in order	 Recommended — no vMotion dependency, simplest/safest resize, leverages the existing Lab Shutdown/Startup Guide

Recommended: Approach B (full maintenance window). Because vMotion can't be relied on, evacuating hosts live isn't dependable. Bringing the whole nested stack down cleanly, resizing all four hosts at once, and starting back up is both **simpler and lower-risk** than a rolling change that hinges on a broken vMotion path — and it reuses the procedure you already have in `VCF_Lab_Shutdown_Startup_Guide`.

7. Procedure — Full Maintenance Window

Follow the existing `VCF_Lab_Shutdown_Startup_Guide` for the exact shutdown/startup order; the rebalance-specific steps are inserted while everything is down.

1. **Pre-flight (§5)** — vSAN healthy, 0 resync, placement noted, DRS Partially Automated.
2. **Graceful shutdown (per the guide), in order:** stop Aria (`/opt/scripts/deploy.sh --shutdown` then power off), then the other management VMs (vcf-ops, fleet, vrsldm, idm, sddc-manager, nsx-manager, **vcenter last** of the appliances), then **vCLS** is handled by the cluster.
3. **vSAN cluster shutdown** — use the supported **vSAN cluster shutdown** workflow (or, with vCenter down, the per-host `esxcli system maintenanceMode + python /usr/lib/vmware/vsan/bin/reboot_helper.py prepare` path) so vSAN is cleanly quiesced. *Verify the exact VCF 9 shutdown step before running.*
4. **Power off the four nested ESXi VMs** in VMware Workstation.
5. **Resize in Workstation** (per §4): set each nested ESXi VM's RAM and vCPU to target. Double-check the **sum of RAM ≤ 352 GB**.

6. **Power on the nested ESXi VMs**, let vSAN come up, **exit maintenance mode**, confirm vSAN network + disks healthy.
7. **Start the management stack (per the guide), in order:** vcenter first, then sddc-manager, nsx-manager, vcf-ops, fleet, vrsbcm, idm; Aria last.
8. **Set placement (§4):** with DRS Partially Automated, **power each heavy VM onto its intended host** (or create **VM-Host "should-run" rules**) so `aria / nsx-manager / vcenter` land on esxi04/esxi01/esxi03 respectively.
9. **Re-enable / reconfigure HA** if it was disabled; let admission-control recalculate (the reserve should drop now that hosts are more even).

8. Verification

- **vSAN:** object health summary clean, **0 inaccessible**, resync drained to 0.
- **Hosts:** each host RAM% at rest matches §4 (~33–60%), none hot; HA all `connectedToMaster / master`.
- **HA reserve:** admission-control reserved % dropped from 34/25 toward an even ~25/25.
- **Placement:** `aria / nsx-manager / vcenter` on separate hosts as intended.
- **Workloads:** vCenter, NSX, VCF Operations, Aria all back and healthy.
- **DRS:** left **Partially Automated** (do not return to Fully Automated until storage/vMotion is healthy).

9. Risk & Rollback

Risk	Mitigation
Total RAM exceeds 384 → Windows host starves / Workstation swaps	Keep nested total ≤ 352; verify the arithmetic in step 5
Host sized too small → VM won't power on / vSAN starved	Validate targets in §5; floor ≈ 64 GB per heavy-VM host
vSAN doesn't come back cleanly	Clean vSAN shutdown (step 3); verify health before powering VMs
Startup-order mistake	Follow <code>vcf-Lab-Shutdown-Startup-Guide</code> exactly (vcenter early, Aria last, lcm/Fleet per its notes)
Something wrong after resize	Rollback: power off nested hosts, restore the previous RAM/vCPU values (recorded in §5), power back up

Rollback is simple because the only change to the hosts is RAM/vCPU numbers in Workstation — revert the numbers and restart. No data is moved or destroyed.

10. The Decision

Decision: Rebalance the four nested hosts to **esxi01 96 / esxi02 72 / esxi03 88 / esxi04 96 GB** (total 352, unchanged), trim esxi04 to 24 vCPU, and **redistribute the heavy VMs so aria / nsx-manager / vcenter sit on three separate hosts** — executed in a **full maintenance window** (Approach B). **Justification:** Relieves the 93%-hot esxi03, evens the load, shrinks the HA reserve, and removes the imbalance that triggered the DRS/HA cascade — without depending on the currently-unreliable vMotion, and within the 384 GB ceiling. **Implication:** Requires a full management-stack maintenance window

(vCenter UI down during the work); DRS stays Partially Automated and placement is set manually (or via should-run rules); exact targets must be validated against real vSAN overhead first.

11. Document Information

Field	Value
Title	Nested ESXi Host CPU/RAM Rebalance Plan
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Related Documents	VCF_Lab_Shutdown_Startup_Guide; VCF9-SDDC-Manager-Inventory-State-Known-Issues-2026-06-08; PLAN-Post-vCenter-Redeploy-Remediation-SDDC-NSX-Console-2026-06-05
Distribution	Lab notebook

Revision History

Version	Date	Notes
1.0	2026-06-08	Initial plan. Goal, measured current state, hard constraints (384 GB ceiling, vSAN overhead, resize-requires-power-off, vMotion unreliable, vSAN data availability), proposed host targets + heavy-VM placement (aria/nsx/vcenter on three hosts), read-only pre-flight checks, the rolling-vs-full-window comparison (full window recommended because vMotion is unreliable), the maintenance-window procedure, verification, risk/rollback, and the design decision. No changes made; targets to be validated before execution.
1.1	2026-06-08	Post-execution update. Plan executed via Approach B; host resize applied as designed. Added a top-of-doc update note and superseded Constraint #4: vMotion verified working again during bring-up (idm/vrslcm live-migrated past the old ~56% stall, vSAN resync 0). Noted DRS-auto-placement scattered the light VMs onto esxi04 on power-on (corrected via live vMotion + soft should-run rules — see the Execution Worksheet). DRS kept Partially Automated pending a longer reliability window.

Document End

See also: - *VCF_Lab_Shutdown_Startup_Guide* — the exact shutdown/startup order this plan relies on - *VCF9-SDDC-Manager-Inventory-State-Known-Issues-2026-06-08* — the accepted inventory items; this rebalance removes the DRS pressure that exposed them